Julien HORGUEDEBAT

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Presentation

Student in the Master program in Advanced Mathematics at the Université Marie et Louis Pasteur in Besançon (France)

Education

Université Marie et Louis Pasteur *Sept 2023 – present*
Master degree in Advanced Mathematics

- **Coursework:** Numerical Analysis, Partial and Ordinary Differential Equations, Student Seminar, Fields Theory, Algebra, Complex Analysis, Fourier Analysis, Topology, Probabilities

Université Marie et Louis Pasteur *Sept 2022 – June 2023*
Bachelor degree in Fundamental Mathematics

- Result: 15.415/20
- **Coursework:** Algebra, Metric spaces, Integration Theory, Functional Analysis, Probabilities, Affine Structure, Numerical Analysis, Differential calculus

Université Marie et Louis Pasteur *Sept 2022 – present*
Magistère in Mathematics

- **Coursework:** Epistemology, Statistics, Functional Analysis, Introduction to C, Colloquium

Experience

Research Internship *Besançon, FR*
Laboratoire de Mathématiques de Besançon - UMR 6623 *May 2024 – Jul 2024*

- Implementation of numerical methods to solve integral equations using the Nyström method.

Handler *Besançon, FR*
Grand Besançon Métropole *June 2023*

Seminars Attended

Laboratoire de Mathématiques de Besançon - UMR 6623 *Sept 2025 – present*
Numerical Analysis and Scientific Computing

- Weekly attending in the workgroup and seminar of the Numerical Analysis and Scientific Computing team of the laboratory.

Research Initiation School of the LmB *Jan 2023*
Winter School of the Maths Master of LmB *Jan 2025*
Laboratoire de Mathématiques de Besançon - UMR 6623 *Dec 2025*

- Conferences by members of the laboratory of mathematics

Lectures Sophie Kowalevski *4-6 June 2024*
Centre Henri Lebesgue - Angers

- Short-courses in algebra and analysis/probabilities.

Academic Projects

Moment Problems *Oct 2025 – present*
Université Marie et Louis Pasteur; directed by Thierry DAUDE and Bastien POLLIZI

- Methods of resolution for moment problems and numerical implementation
- Tools: Julia, LaTeX

Master 1 thesis: Fourier Transform on a Finite Commutative Group *Jan 2024 – May 2024*
Université Marie et Louis Pasteur; directed by Cécile ARMANA; group of 2

- Project about group theory, Fourier transform on finite commutative groups, Fast Fourier Transform and applications to polynomial multiplication.
- Tools: Python, LaTeX

Bachelor Project: On the existence of invariant measures under the action of a group and the Banach-Tarski paradox *Jan 2023 – May 2023*
Université Marie et Louis Pasteur; directed by Marsault CHABAT; group of 3

- Project about group theory, the Banach-Tarski paradox and the existence of non-measurable sets in $\mathbb{R}$.
- Tools: LaTeX

Skills

Programming languages: Python (Advanced), Julia (Advanced), LaTeX(Advanced), C (basic)
Languages: French (Mother Tongue), English (Technical), Spanish (Basic)
First aid certification (PSC1)